

Tabular Planning: Policy Evaluation

**Kianté Brantley
CS 1840 Fall 2026**

Announcements

HW1 is out today

HW0 is due Sept 17 (**this Thursday**) , 11:59 pm ET

Check the course website for Office Hours:
<https://harvard-cs-1840.github.io/2026/>

Recap: Definitions

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

$$P : S \times A \mapsto \Delta(S), \quad r : S \times A \rightarrow [0,1], \quad \gamma \in [0,1)$$

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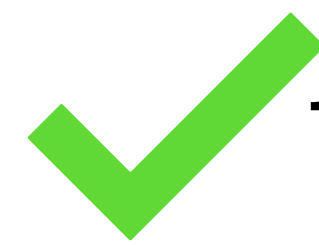
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$$\text{Policy } \pi : S \mapsto A$$

$$\text{Value function } V^\pi(s) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \mid s_0 = s, a_h = \pi(s_h), s_{h+1} \sim P(\cdot \mid s_h, a_h) \right]$$

Finishing Up From Last Lecture



1. Definitions of Markov Decision Process

2. Value functions (V and Q functions)

3. Bellman equations

State-action Q function

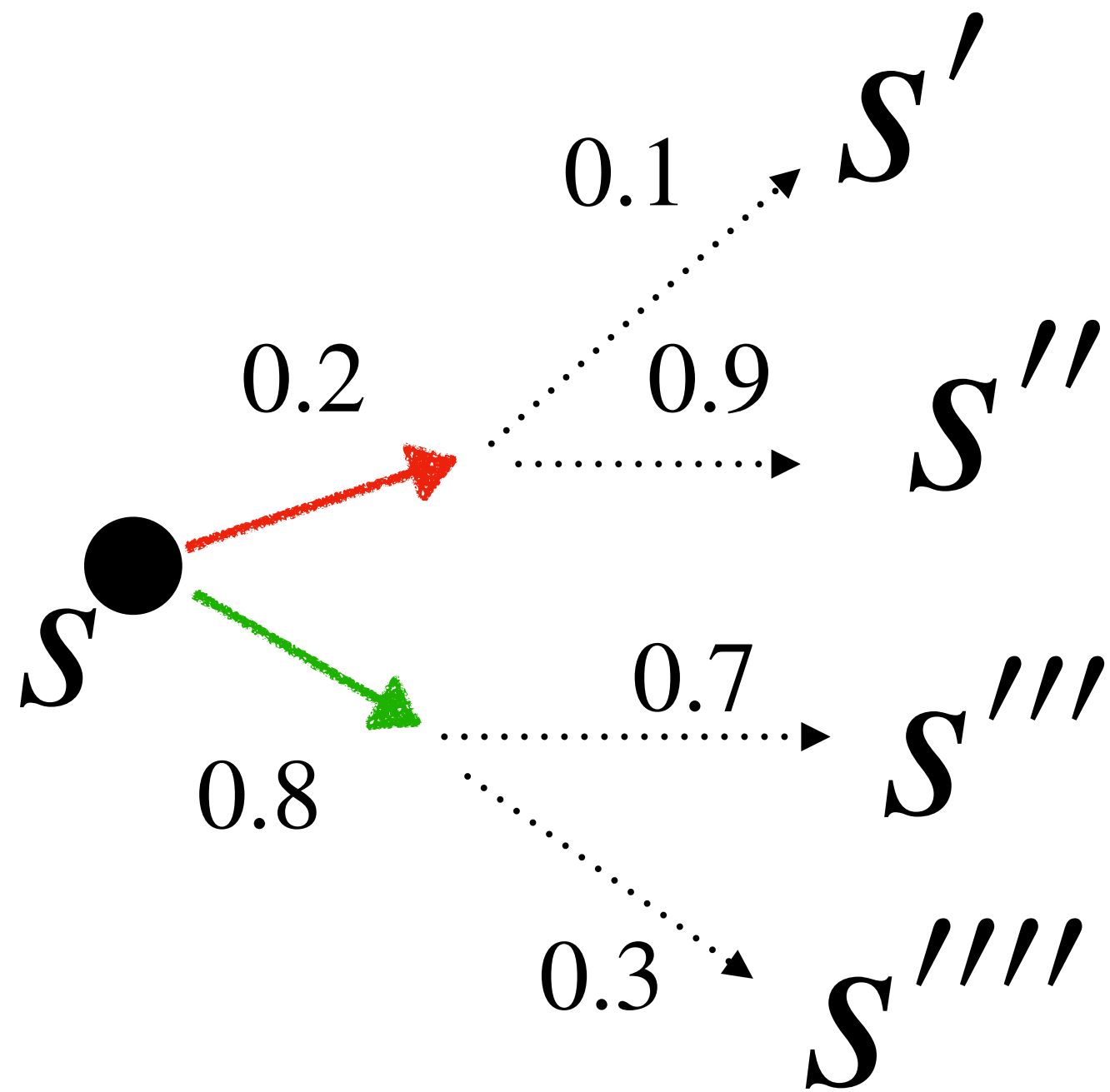
$$Q^{\pi}(s, a) = \mathbb{E} \left[r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a_0 = a, \pi \right]$$

Can we express the value function using only one step?

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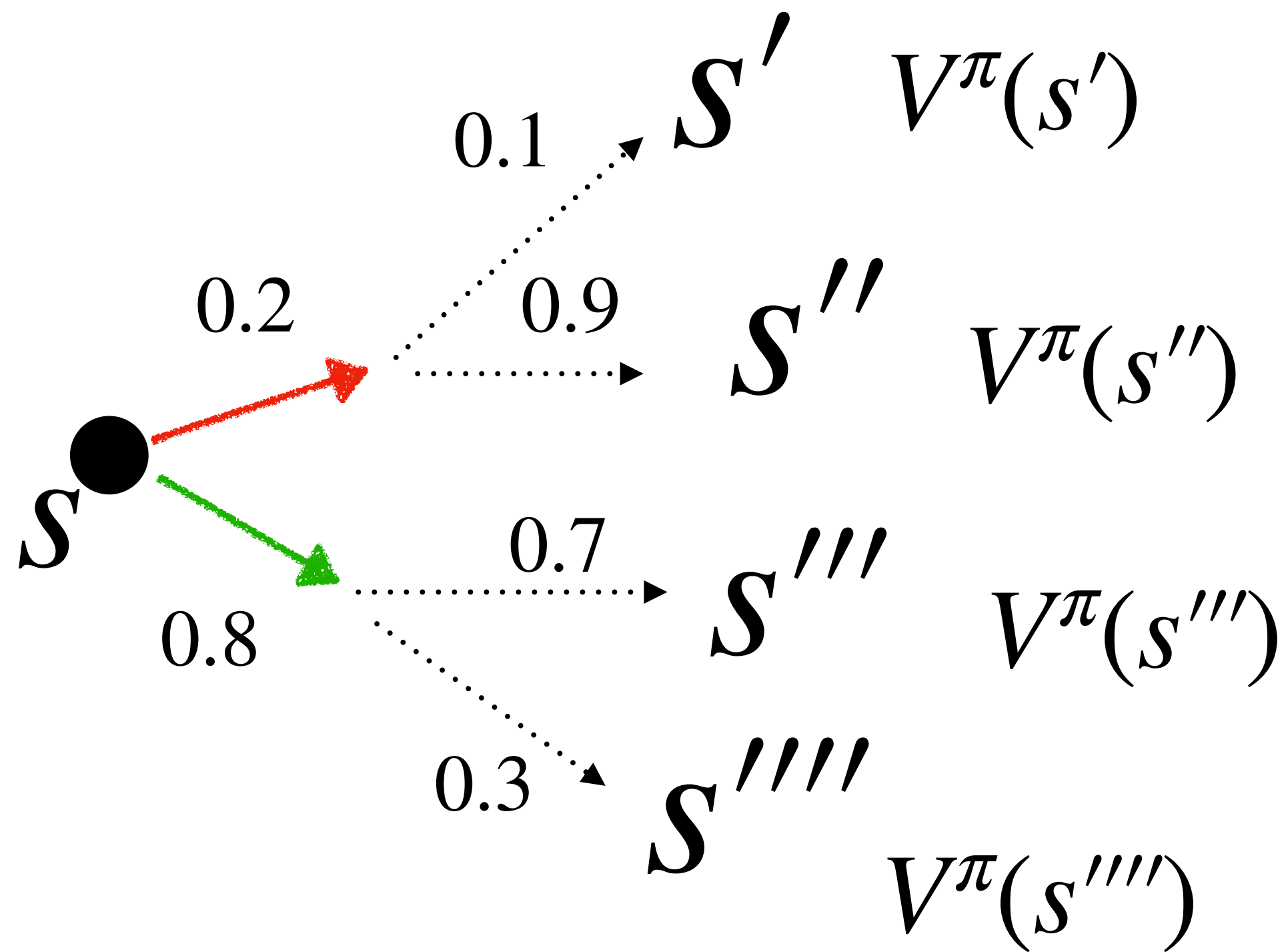
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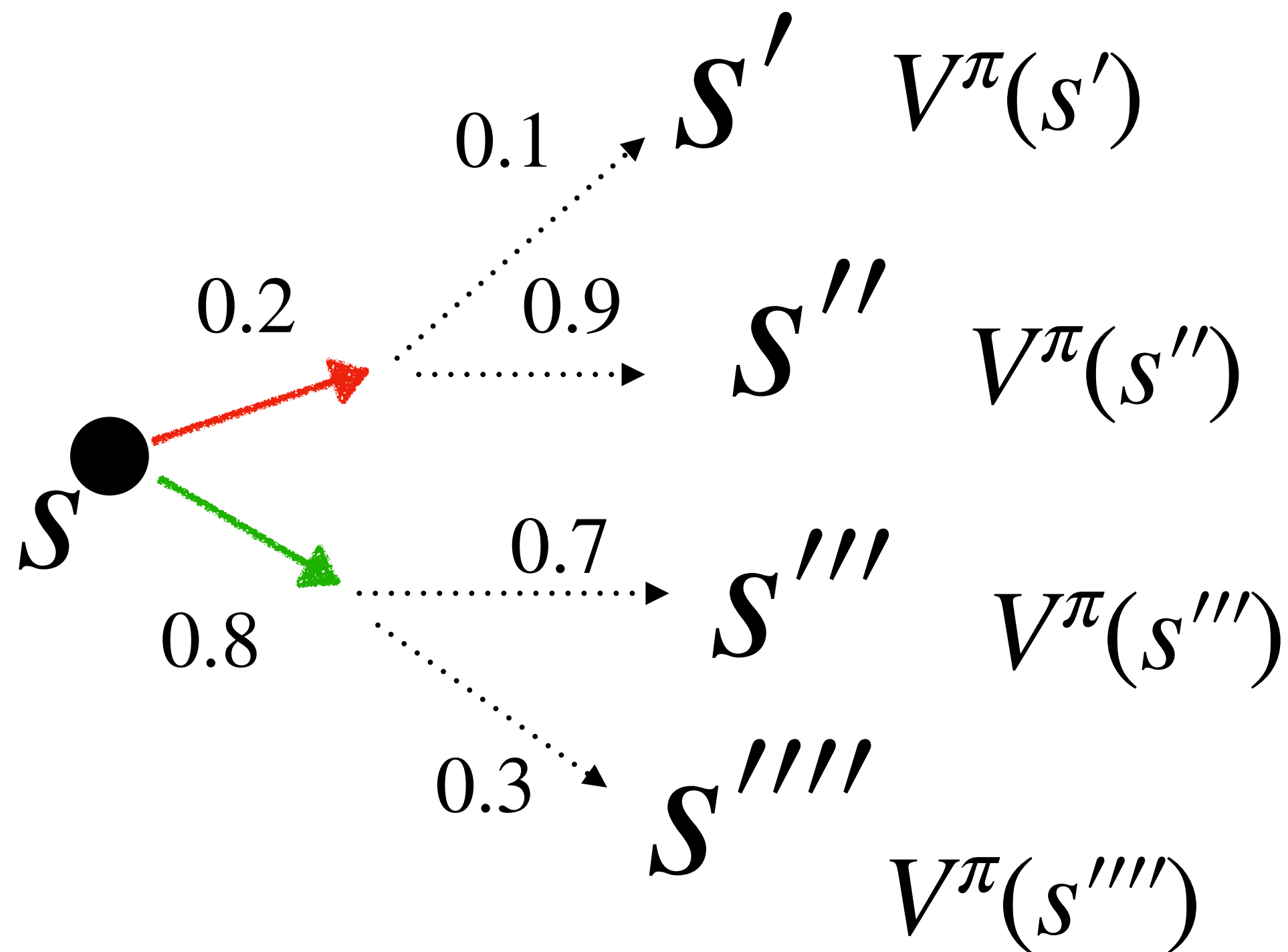
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$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^\pi(s')] \right]$$

Bellman equation for value function

Bellman equation (consistency)

Bellman Eq

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^\pi(s')] \right], \forall s$$

Relationship between V and Q

Exercise: Can you express V^π in terms of Q^π , and Q^π in terms of V^π ?

Today: Policy Evaluation

Key Question:

**Given MDP $\mathcal{M} = (S, A, r, P, \gamma)$ and a $\pi : S \mapsto A$,
how good is π ?**

i.e., how to compute $V^\pi(s), \forall s$?

Motivation for Policy Evaluation



We want to **evaluate** our strategy against some opponent (we can abstract our strategy as policy π)

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We want to **evaluate** our recommendation strategy before we release it to users

A more fundamental motivation...

Recall that we have A^S many policies.
To select the optimal policy, we need to do evaluation

Outline:

1. **Exact** Policy Evaluation

2. **Approximate** Policy Evaluation via an Iterative Algorithm

Exact Policy Evaluation

Setup: we have MDP $\mathcal{M} = (S, A, P, \gamma, r)$, and deterministic policy π ; we want to compute V^π

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This gives us S many linear constraints

Exact Policy Evaluation

Let's form linear constraints. Denote $V(s)$ as our estimator for $s \in \mathcal{S}$

$$\forall s, V(s) = r(s, \pi(s)) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, \pi(s)) V(s')$$

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$$\forall s, V(s) = r(s, \pi(s)) + \gamma \sum_{s' \in S} P(s' | s, \pi(s)) V(s')$$

Denote $V \in \mathbb{R}^{|S|}$, $R \in \mathbb{R}^{|S|}$, where $R_s = r(s, \pi(s))$, and
 $P \in \mathbb{R}^{|S| \times |S|}$, where $P_{s,s'} = P(s' | s, \pi(s))$,

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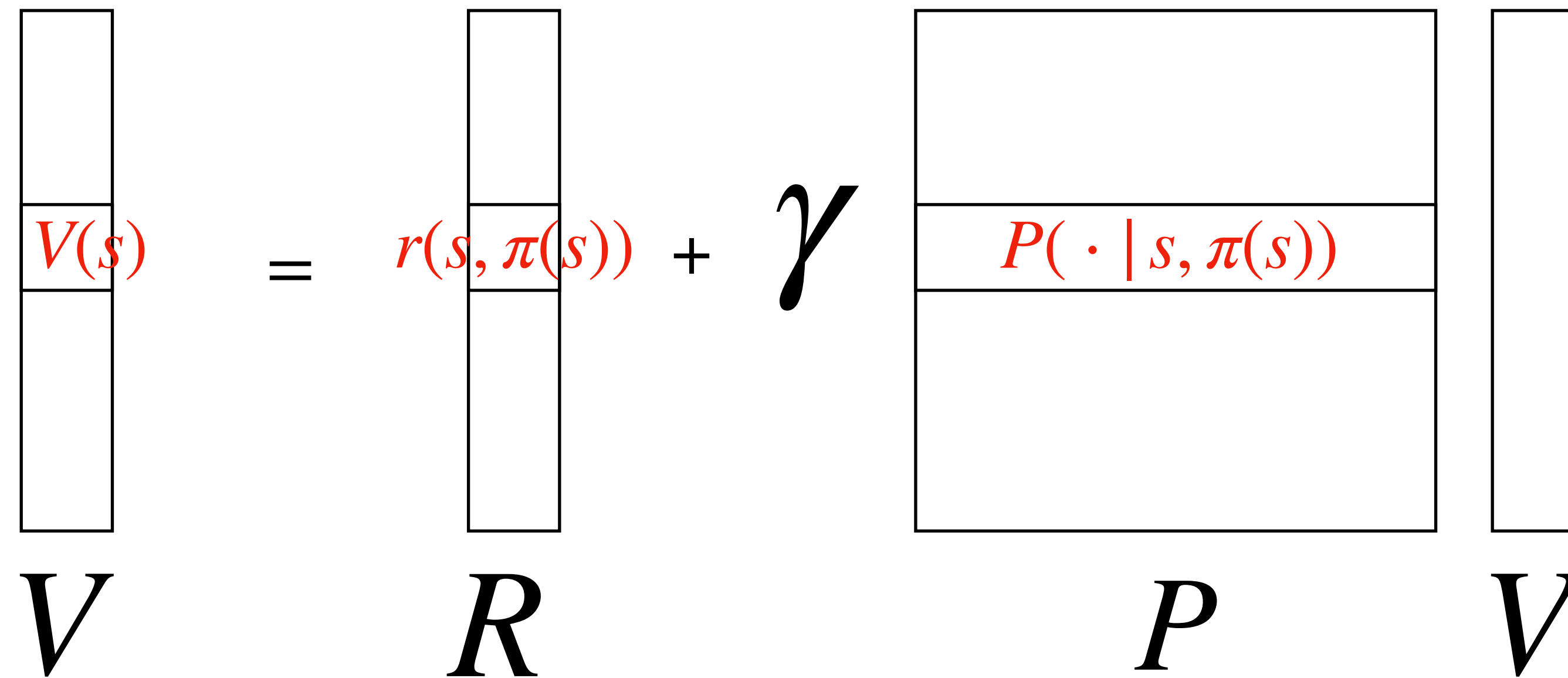
we can **combine all S many constraints together:**

$$V = R + \gamma PV$$

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Since $V = R + \gamma P V$, we can obtain V as:

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$$V = (I - \gamma P)^{-1} R$$

See the assigned AJKS section for proving $(I - \gamma P)$ is full rank (thus invertible)

Summary so far:

$$\begin{array}{c} \boxed{} \\ \boxed{V(s)} \\ \boxed{} \end{array} = \begin{array}{c} \boxed{} \\ \boxed{r(s, \pi(s))} \\ \boxed{} \end{array} + \gamma \begin{array}{c} \boxed{} \\ \boxed{P(\cdot | s, \pi(s))} \\ \boxed{} \end{array} \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \end{array}$$

$V \qquad R \qquad P \qquad V$

$$V = (I - \gamma P)^{-1} R$$

Summary so far:

The diagram illustrates the Bellman optimality equation $V = R + \gamma P V$ using rectangular blocks. On the left, a vertical rectangle labeled V has its middle section labeled $V(s)$. This is followed by an equals sign. To the right of the equals sign is a vertical rectangle labeled R with its middle section labeled $r(s, \pi(s))$. This is followed by a plus sign and a large italicized Greek letter γ . To the right of γ is a large square labeled P with its middle section labeled $P(\cdot | s, \pi(s))$. Finally, to the right of the square is a vertical rectangle labeled V .

$$V = (I - \gamma P)^{-1} R$$

Downside: computationally expensive; matrix inverse is $O(S^3)$

Outline:

 1. Exact Policy Evaluation

2. Approximate Policy Evaluation via an Iterative Algorithm

(An approximation solution could be enough, i.e., trade accuracy for computation)

Detour: fix-point solution

Consider $x^\star = f(x^\star)$, $f : [a, b] \mapsto [a, b]$

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Common approach to find $x^\star$:

Initialize $x^0 \in [a, b]$, repeat: $x^{t+1} = f(x^t)$

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If f is a contraction mapping,

i.e., $\forall x, x', |f(x) - f(x')| \leq \gamma |x - x'|$, for some $\gamma \in [0, 1)$, then:

$x^t \rightarrow x^\star$, as $t \rightarrow \infty$

V^π is a fix-point solution:

$$\forall s, V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$

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$$V^\pi = R + \gamma P V^\pi$$

Iterative Policy Evaluation:

Algorithm (Iterative PE):

Start with some initialization $V^0 \in [0, 1/(1 - \gamma)]^{|S|}$, repeat for $t = 0 \dots$:

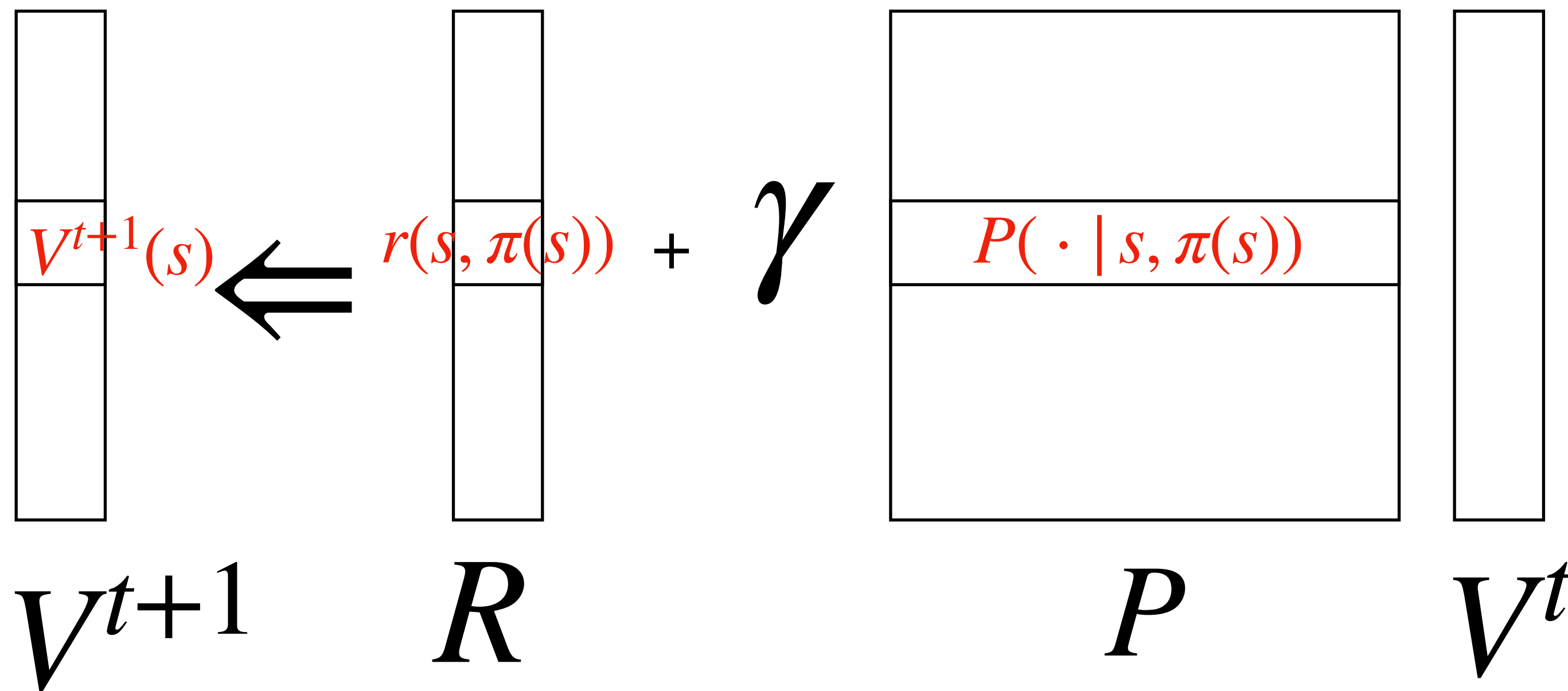
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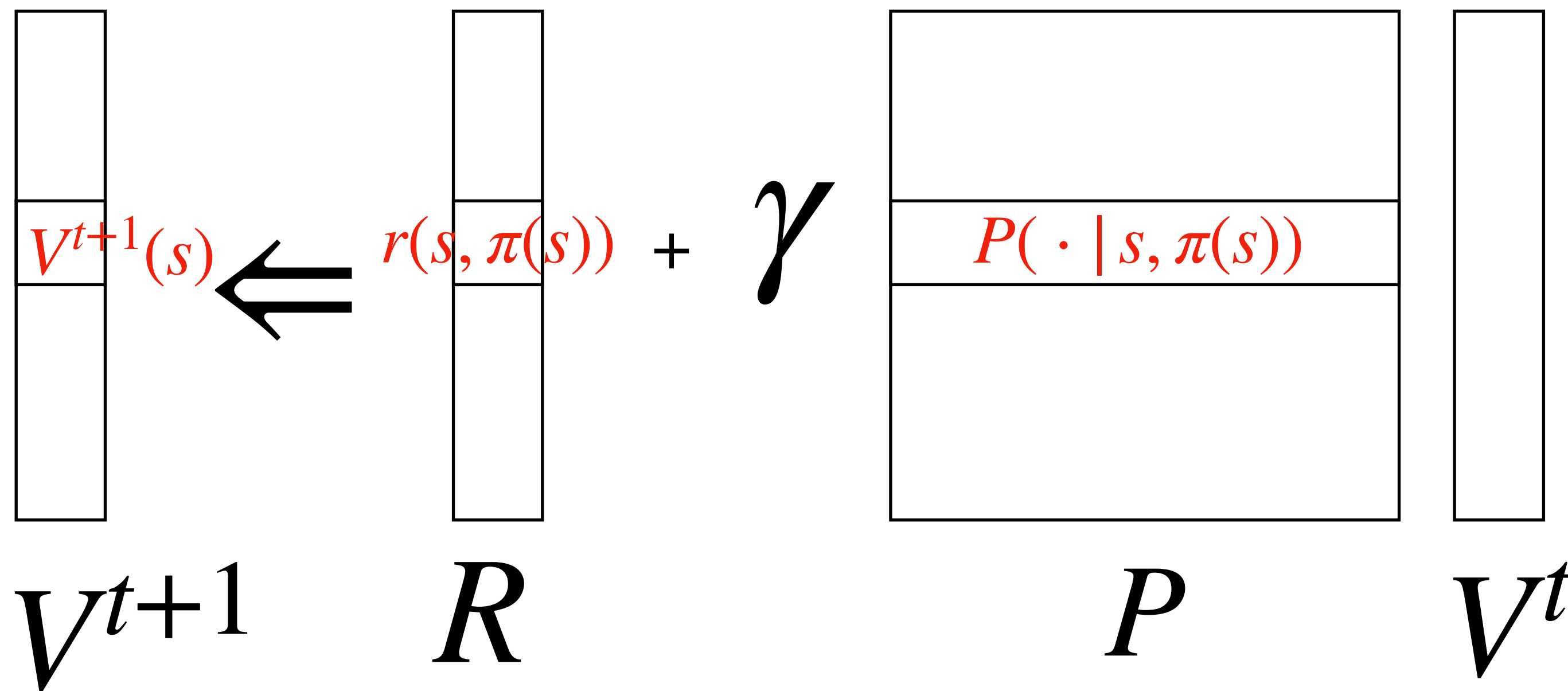


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Q: What's computation complexity per iteration?

Iterative Policy Evaluation:

Why does this iterative process makes sense?

$$V^{t+1} \Leftarrow R + \gamma P V^t$$

$$V^{t+1}(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^t(s')$$

$$V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0,1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

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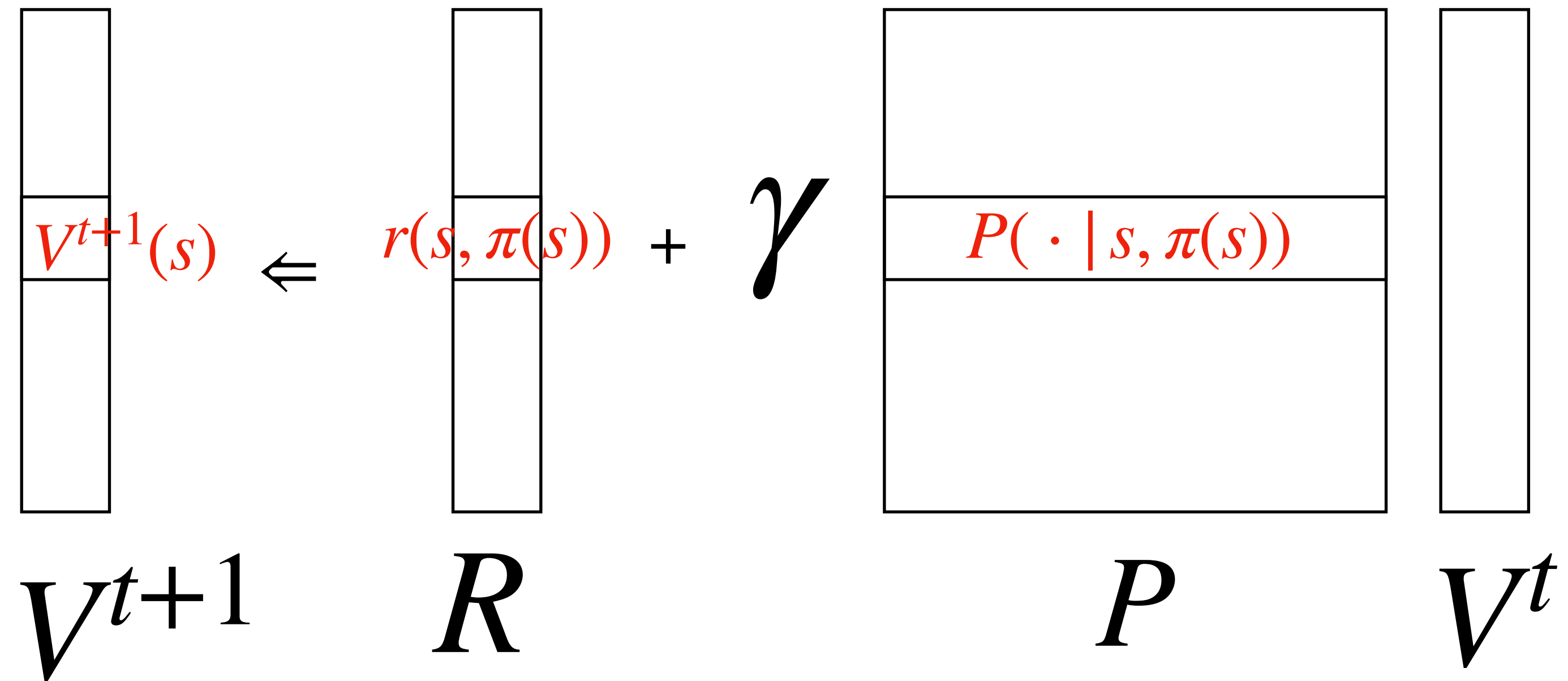
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$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|$$

$$\leq \gamma \left\| V^t - V^\pi \right\|_\infty \Rightarrow \left\| V^{t+1} - V^\pi \right\|_\infty \leq \gamma \left\| V^t - V^\pi \right\|_\infty$$

Summary so far:



Convergence:

$$\| V^{t+1} - V^\pi \|_\infty \leq \gamma \| V^t - V^\pi \|_\infty \leq \gamma^{t+1} \| V^0 - V^\pi \|_\infty$$

Outline:

-  1. Exact Policy Evaluation
-  2. Approximate Policy Evaluation via an Iterative Algorithm

Extension to stochastic policy and Q functions

How would we modify exact and iterative policy evaluation to:

1. evaluate a stochastic policy $\pi(a \mid s)$?
2. compute $Q^\pi(s, a)$ instead of $V^\pi(s)$?

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1. The **exact** algorithm requires the matrix $V = (I - \gamma P)^{-1}R$ inverse (computation complexity at least) $O(S^3)$

Summary

Key Question today: Given MDP $\mathcal{M}$ and a policy π , How to compute $V(s)$, $\forall s$?

1. The **exact** algorithm requires the matrix $V = (I - \gamma P)^{-1}R$ inverse (computation complexity at least) $O(S^3)$
2. An iterative algorithm can quickly find an **approximate** solution (error shrinks at a rate of γ^t)

Summary

Feedback

<https://bit.ly/4gYn2nC>

